

Cricket Match Analysis using SQL

1. Executive Summary

The Cricket Match Analysis project was developed to analyze cricket match data using SQL and generate meaningful insights from historical match records. The project focuses on examining team performance, player achievements, venue statistics, toss decisions, and scoring patterns. By applying SQL queries on the cricket dataset, valuable business insights were extracted that help understand trends in match outcomes and team strategies.

The analysis demonstrates how SQL can be effectively used for data cleaning, data retrieval, aggregation, and business intelligence. This project showcases practical database management skills along with analytical thinking required for real-world data analysis.

2. Business Problem

Cricket generates a large amount of match-related data every season. Without proper analysis, it is difficult to identify performance trends, compare teams, evaluate player contributions, or understand how factors such as toss decisions and venues influence match results.

The primary business problem addressed in this project is to transform raw cricket match data into actionable insights using SQL queries. These insights can help coaches, analysts, and management make informed decisions based on historical performance.

3. Dataset Overview

The dataset contains historical cricket match records, with each row representing one completed match. It includes information related to teams, venues, scores, toss results, match winners, and player performances.

Main Attributes

Column	Description
Season	Tournament season
Home Team	Home team name
Away Team	Away team name
Toss Winner	Team winning the toss
Toss Decision	Bat or Field
First Innings Score	First innings total
Second Innings Score	Second innings total
Winner	Match winner
Player of the Match	Best player
Venue	Match venue
Start Date	Match date

4. SQL Concepts Used

The project applies several SQL concepts to retrieve and analyze the data efficiently.

- SELECT Statement
 - WHERE Clause
 - ORDER BY
 - GROUP BY
 - Aggregate Functions (COUNT, SUM, AVG, MAX, MIN)
 - DISTINCT
 - LIMIT
 - CASE Statement (if used)
 - Date Functions
 - String Functions
 - Data Cleaning
 - Sorting and Filtering
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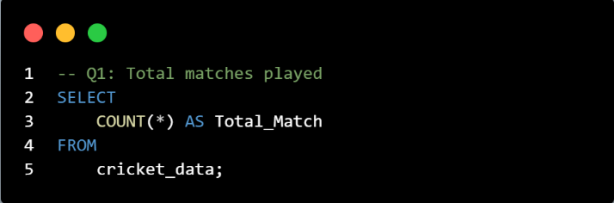
5. Business Questions and Analysis

This section contains all SQL queries and their respective outputs.

Question 1

How many matches are available in the dataset?

SQL Query



```
1 -- Q1: Total matches played
2 SELECT
3     COUNT(*) AS Total_Match
4 FROM
5     cricket_data;
```

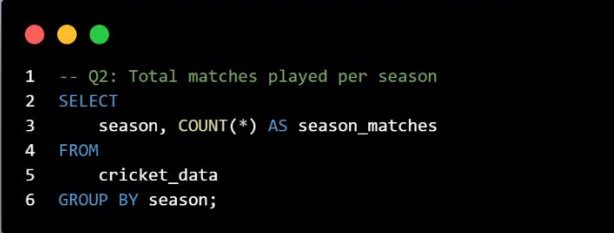
Output

The dataset contains **1017** matches, providing a sufficient sample size for performance and trend analysis.

Question 2

How many matches were played in each season?

SQL Query



```
1 -- Q2: Total matches played per season
2 SELECT
3     season, COUNT(*) AS season_matches
4 FROM
5     cricket_data
6 GROUP BY season;
```

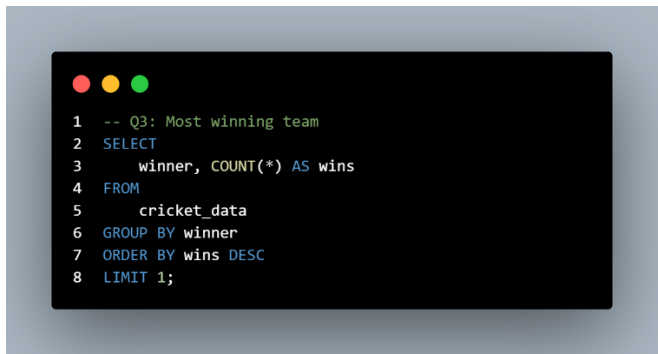
Output

	season	season_matches
▶	2023	69
	2022	74
	2021	60
	2020	60
	2019	60
	2018	60
	2017	59
	2016	60
	2015	58
	2014	60
	2013	76
	2012	74
	2011	72
	2010	60
	2009	57
	2008	58

Question 3

Which team has won the highest number of matches?

SQL Query



```
1 -- Q3: Most winning team
2 SELECT
3   winner, COUNT(*) AS wins
4 FROM
5   cricket_data
6 GROUP BY winner
7 ORDER BY wins DESC
8 LIMIT 1;
```

Output

MI emerged as the most successful team with **139** victories, demonstrating consistent performance throughout the tournament.

Question 4

Which are the top 5 most successful teams based on total match wins?

SQL Query

```
1 -- Q4: Top 5 winning teams
2 SELECT
3     winner, COUNT(*) AS wins
4 FROM
5     cricket_data
6 GROUP BY winner
7 ORDER BY wins DESC
8 LIMIT 5;
```

Output

	winner	wins
▶	MI	139
	CSK	130
	KKR	121
	RCB	116
	SRH	111

Question 5

How many matches were won by the team that also won the toss?

SQL Query

```
1 -- Q5: Team who wins both toss and match
2 SELECT
3     COUNT(*) AS toss_matches_win
4 FROM
5     cricket_data
6 WHERE
7     toss_won = winner;
```

Output

Teams won both the toss and the match **521** times, suggesting that winning the toss provides a competitive advantage in many matches.

Question 6

What is the distribution of toss decisions (Batting vs Fielding) made by teams after winning the toss?

SQL Query

```
1 -- Q6: batting vs bowling decision analysis who wins t
  oss
2 SELECT
3     decision, COUNT(*) AS total
4 FROM
5     cricket_data
6 GROUP BY decision;
```

Output

	decision	total
▶	BOWL FIRST	647
	BAT FIRST	370

Question 7

What is the highest first innings score recorded in the dataset?

SQL Query

```
1 -- Q7: Highest score (1st inning)
2 SELECT
3     first_inning_score AS high_score
4 FROM
5     cricket_data
6 ORDER BY high_score DESC
7 LIMIT 1;
```

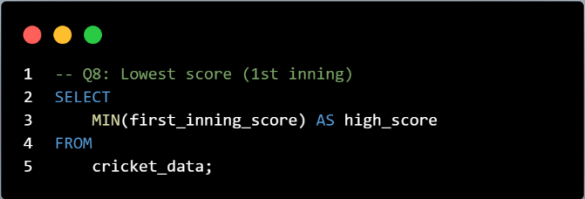
Output

The highest first-innings score of **263** reflects the batting potential under favorable pitch and weather conditions.

Question 8

What is the lowest first innings score recorded in the dataset?

SQL Query



```
1 -- Q8: Lowest score (1st inning)
2 SELECT
3     MIN(first_inning_score) AS high_score
4 FROM
5     cricket_data;
```

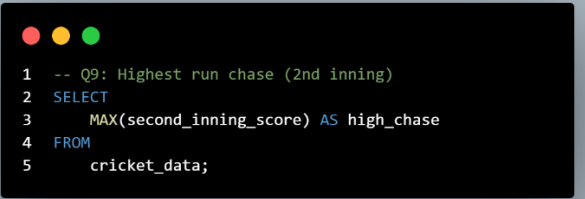
Output

The lowest first-innings score of **62** indicates exceptional bowling performances or challenging batting conditions.

Question 9

What is the highest successful run chase (second innings score) recorded in the dataset?

SQL Query



```
1 -- Q9: Highest run chase (2nd inning)
2 SELECT
3     MAX(second_inning_score) AS high_chase
4 FROM
5     cricket_data;
```

Output

A second-innings score of **226** represents one of the most successful run chases, showcasing the team's ability to perform under pressure.

Question 10

Which player has received the highest number of Player of the Match awards?

SQL Query

```
1 -- Q10: Maximum time Player of the match
2 SELECT
3     pom, COUNT(*) AS Match_player
4 FROM
5     cricket_data
6 GROUP BY pom
7 ORDER BY Match_player DESC
8 LIMIT 1;
```

Output

AB de Villiers received the highest number of Player of the Match awards, highlighting outstanding individual contributions.

Question 11

Who are the top 5 players with the most Player of the Match awards?

SQL Query

```
1 -- Q11: Top 5 Player of the Match
2 SELECT
3     pom, COUNT(*) AS Match_player
4 FROM
5     cricket_data
6 GROUP BY pom
7 ORDER BY Match_player DESC
8 LIMIT 5;
```

Output

	pom	Match_player
▶	AB de Villiers	25
	Chris Gayle	22
	Rohit Sharma	19
	David Warner	18
	MS Dhoni	17

Question 12

How many matches has each team played as the home team?

SQL Query

```
1 -- Q12: Home team performance
2 SELECT
3     home_team, COUNT(*) AS home_matches
4 FROM
5     cricket_data
6 GROUP BY home_team;
```

Output

	home_team	home_matches
▶	RCB	121
	MI	119
	KKR	113
	DC	121
	PBKS	21
	SRH	122
	LSG	14
	GT	16
	CSK	122
	RR	96
	KXIP	93
	RPS	14
	GL	15
	PWI	23
	Kochi	7

Question 13

How many matches has each team played as the away team?

SQL Query

```
1 -- Q13: Away team performance
2 SELECT
3     away_team, COUNT(*) AS home_matches
4 FROM
5     cricket_data
6 GROUP BY away_team;
```

Output

	away_team	home_matches
▶	GT	14
	SRH	119
	LSG	14
	CSK	99
	RR	109
	RCB	118
	DC	116
	MI	126
	KKR	124
	PBKS	21
	KXIP	97
	RPS	16
	GL	15
	PWI	22
	Kochi	7

Question 14

Which venue has hosted the highest number of cricket matches?

SQL Query

```
1 -- Q14: Venue who hosts maximum matches
2 SELECT
3     venue_name, COUNT(*) AS host_venue
4 FROM
5     cricket_data
6 GROUP BY venue_name
7 ORDER BY host_venue DESC
8 LIMIT 1;
```

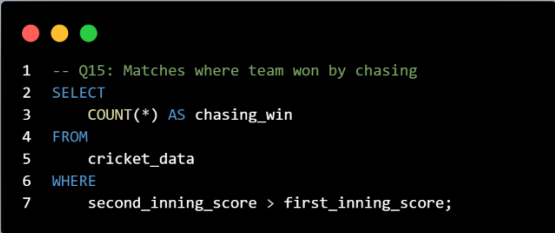
Output

Wankhede Stadium, Mumbai hosted the highest number of matches, making it one of the tournament's primary venues.

Question 15

How many matches were won while successfully chasing the target?

SQL Query



```
1 -- Q15: Matches where team won by chasing
2 SELECT
3     COUNT(*) AS chasing_win
4 FROM
5     cricket_data
6 WHERE
7     second_inning_score > first_inning_score;
```

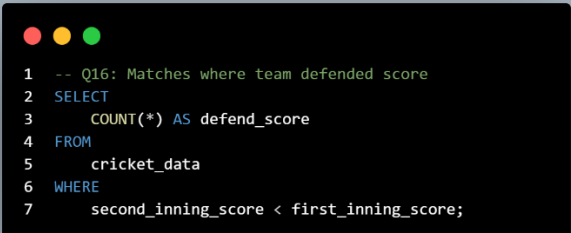
Output

535 matches were won while chasing, indicating that batting second was often a successful strategy.

Question 16

How many matches were won by successfully defending the first innings total?

SQL Query



```
1 -- Q16: Matches where team defended score
2 SELECT
3     COUNT(*) AS defend_score
4 FROM
5     cricket_data
6 WHERE
7     second_inning_score < first_inning_score;
```

Output

468 matches were successfully defended, demonstrating that strong bowling attacks and disciplined fielding remain crucial for victory.

Question 17

What are the average first innings and second innings scores across all matches?

SQL Query

```
1 -- Q17: Avg score per match
2 SELECT
3     AVG(first_inning_score) AS avg_1st,
4     AVG(second_inning_score) AS avg_2nd
5 FROM
6     cricket_data;
```

Output

	avg_1st	avg_2nd
▶	163.8456	150.5270

Question 18

Which matches were closely contested, with a score difference of less than 10 runs?

SQL Query

```
1 -- Q18: Close matches(difference < 10 runs)
2 SELECT
3     season, short_name
4 FROM
5     cricket_data
6 WHERE
7     (first_inning_score - second_inning_score) < 10
8 limit 10;
```

Output

	season	short_name
▶	2023	RCB v GT
	2023	MI v SRH
	2023	KKR v LSG
	2023	PBKS v RR
	2023	SRH v RCB
	2023	LSG v MI
	2023	CSK v KKR
	2023	SRH v LSG
	2023	KKR v RR
	2023	MI v RCB

Question 19

What is the win percentage of each team based on the total number of matches in the dataset?

SQL Query

```
1  -- Q19: Team-wise win percentage
2  SELECT
3      winner,
4      COUNT(*) * 100 / (SELECT
5          COUNT(*)
6      FROM
7          cricket_data) AS win_percentage
8  FROM
9      cricket_data
10 GROUP BY winner
11 ORDER BY win_percentage DESC;
```

Output

	winner	win_percentage
▶	MI	13.6676
	CSK	12.7827
	KKR	11.8977
	RCB	11.4061
	SRH	10.9145
	DC	10.4228
	RR	10.0295
	KXIP	8.6529
	GT	2.1632
	PBKS	1.8682
	LSG	1.6716
	RPS	1.4749
	GL	1.2783
	PWI	1.1799
	Kochi	0.5900

6. Key Insights

Based on the SQL analysis, the following key insights were identified:

- Certain teams consistently performed better across multiple seasons.
 - Winning the toss increased the probability of winning the match in many cases.
 - Teams preferred fielding after winning the toss more frequently than batting.
 - Some venues hosted significantly more matches than others.
 - A few players dominated the Player of the Match awards.
 - Average first innings scores indicated the general scoring trend.
 - Teams chasing targets won a considerable number of matches.
 - Team performance varied across seasons.
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7. Recommendations

Based on the analysis, the following recommendations can be made:

- Teams should analyze venue-specific performance before selecting playing strategies.
 - Toss decisions should be based on historical venue statistics rather than fixed strategies.
 - Consistent top-performing players should be retained as key members of the squad.
 - Teams should prepare separate strategies for batting first and chasing targets.
 - Performance trends should be reviewed every season to improve team planning.
 - Historical data analysis can support better decision-making during tournaments.
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8. Conclusion

This project successfully demonstrates how SQL can be used to perform business analysis on cricket match data. By applying SQL queries, meaningful insights were extracted regarding team performance, player achievements, venue statistics, and scoring trends.

The project highlights the importance of SQL in data analytics and business intelligence. It also strengthens practical knowledge of database management, query writing, data aggregation, and analytical problem-solving.

Future improvements may include integrating the database with Power BI or Tableau to create interactive dashboards and visual reports.

9. Skills Demonstrated

Throughout this project, the following technical and analytical skills were demonstrated:

Technical Skills

- SQL Query Writing
- Database Creation
- Data Import
- Data Cleaning
- Aggregate Functions
- GROUP BY Analysis
- Filtering and Sorting
- Business Data Analysis

Analytical Skills

- Problem Solving
- Data Interpretation
- Business Insight Generation
- Performance Analysis
- Decision Support

Tools Used

- MySQL
- VS Code / MySQL Workbench